6 **(a)** State what is meant by decay constant.

.....

..... [1]

(b) The variation with time t of the number N of technetium-101 nuclei in a sample of radioactive material is shown in Fig. 6.1

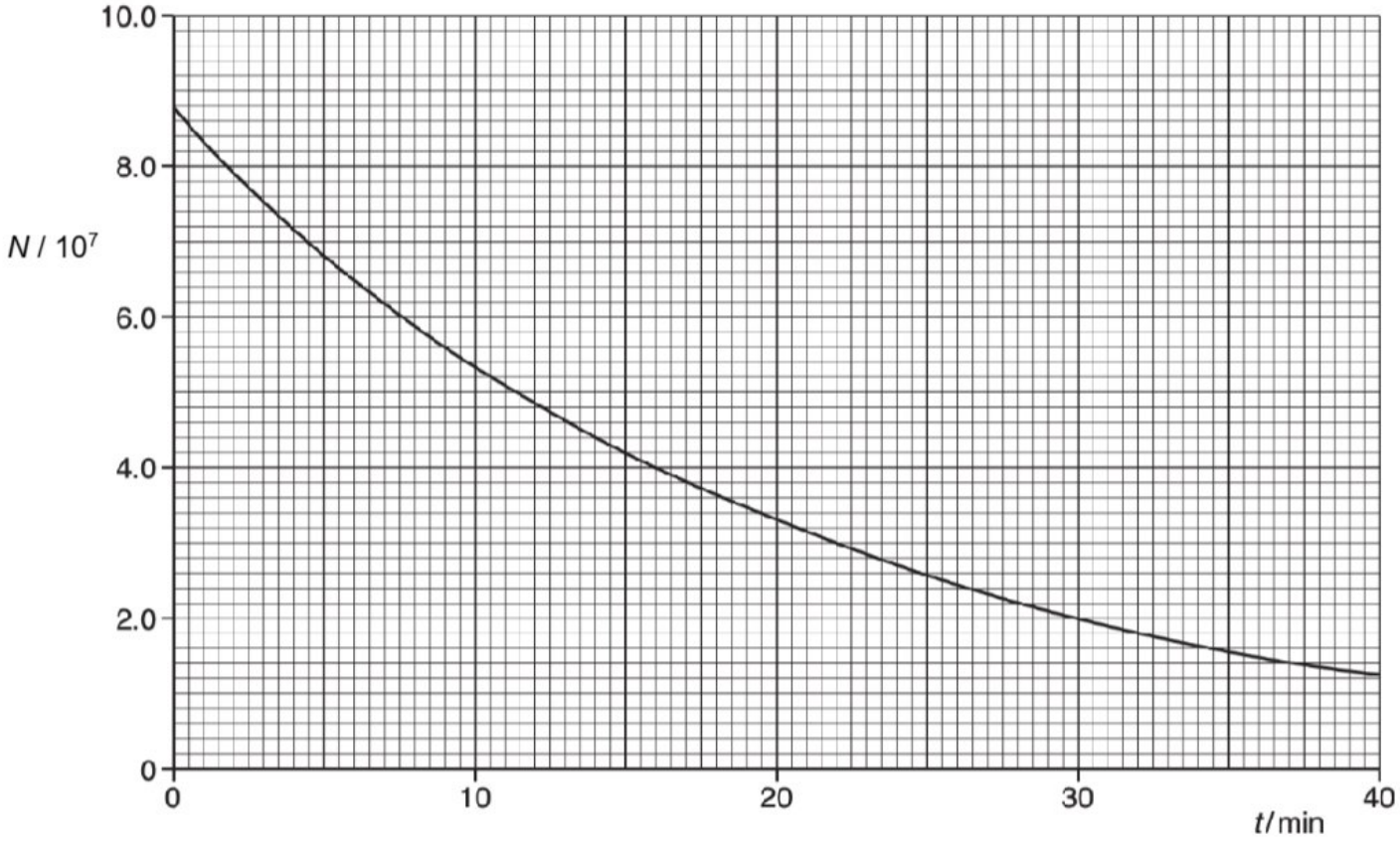


Fig. 6.1

(i) Use Fig. 6.1 to determine the activity, in Bq, of the sample of technetium-101 at the time t = 30 minutes. Show your working clearly.

activity = ... Bq [3]

(ii) Technetium-101 undergoes beta decay into stable Ruthenium.

While beta emission is widely used in targeted cancer therapy, with reference to half-life of technetium-101, suggest why technetium may *not* be suitable for this use.

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..... [1]